

Project 4

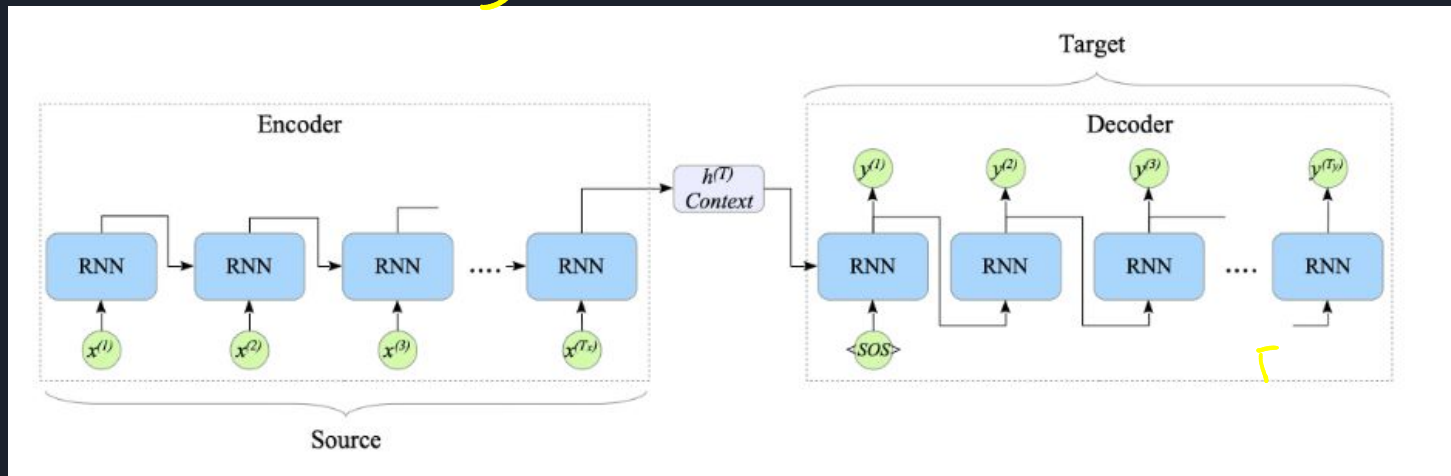
Attention and Transformer

Anshuman Dangwal
The ML Hub

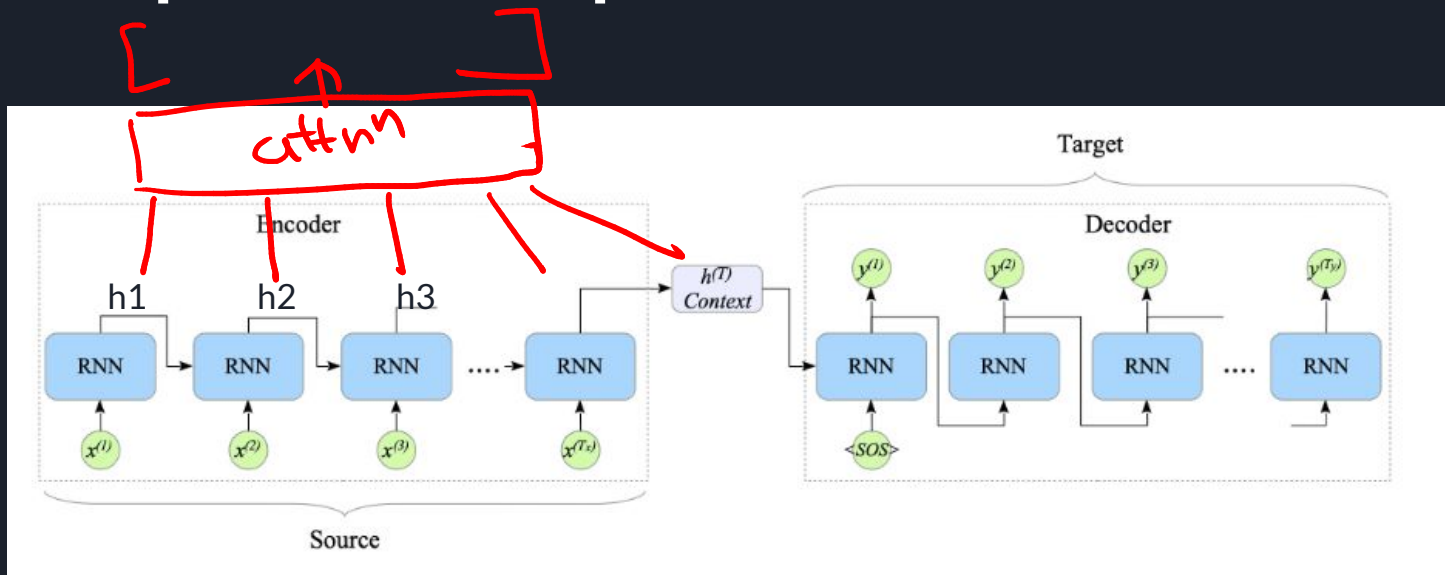
Sequence to Sequence

Examples: Machine Translation, Text Summarization, Audio to Text, etc.

Input seq $\rightarrow$ Encoder $\rightarrow$ latent vector $\rightarrow$ Decoder $\rightarrow$ output seq



Sequence to Sequence



IDEA: Output at every time step is a function of previous output and a linear combination of all hidden states

$$y_i = F(y_{i-1}, \sum_{j=1}^T \alpha_j h_j)$$

Attention

X be input matrix, $X \in \mathbb{R}^{n \times d}$
Learn 3 linear proj.ⁿ of X

$$Q = X W_Q^{d \times d_k} \leftarrow \text{w's are learnable}$$

$$K = X W_K$$

$$V = X W_V$$

$$\underbrace{\quad}_{\leftarrow} \\ n \times d_k$$

Query, Key, Value

Attention

(2)

Take one row of Q and do the following to get its "attention scores"

$$S_i = q \cdot K_i^T \quad \forall K \in K \quad i = 1 \dots N$$

$$S = q \cdot K^T \leftarrow \begin{array}{l} N\text{-length vector} \\ \text{attn score with all key} \end{array}$$

$\downarrow$ $\forall$ rows in Q

$$S = QK^T \rightarrow \hat{S} = \frac{QK^T}{\sqrt{d_k}}$$

$$\alpha = \text{softmax}(\hat{S})$$

Attention

Post attention vector

$$A_q = \sum_{i=1}^n \alpha_i V_i$$

$$\text{softmax}\left(\frac{Q K^T}{\sqrt{d_k}}\right) V$$

Multi-head Attention

$$Q \in \mathbb{R}^{N \times d_k}$$

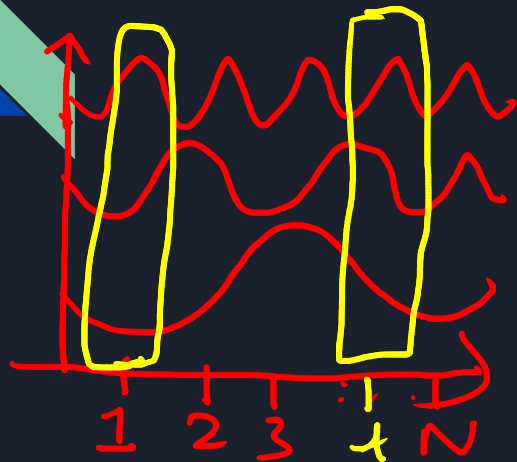
$$N \times \frac{d_k}{m}$$

Learn Multiple Projections of X and concatenate them to get final embeddings

$$Z_1, Z_2, \dots, Z_m, \quad Z_i: \text{Attn}(X)$$

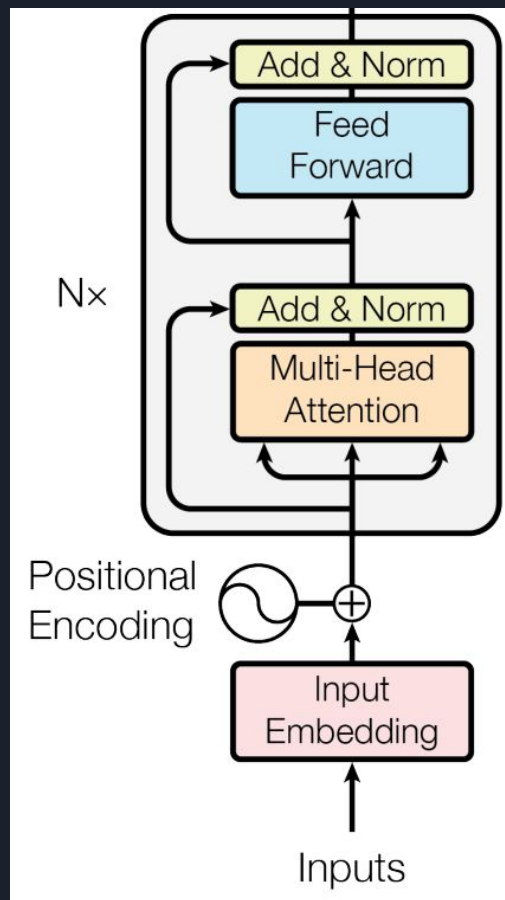
$$\begin{array}{ccc}
 \text{MHA}(X) = [Z_1, Z_2, \dots, Z_m] W_Z & & \\
 N \times d_k & \begin{array}{c} \downarrow \\ N \times \frac{d_k}{m} \end{array} & \begin{array}{c} \downarrow \\ d_k \times d_k \end{array} \\
 & \underbrace{\hspace{10em}} & \\
 & N \times d_k &
 \end{array}$$

Transformer



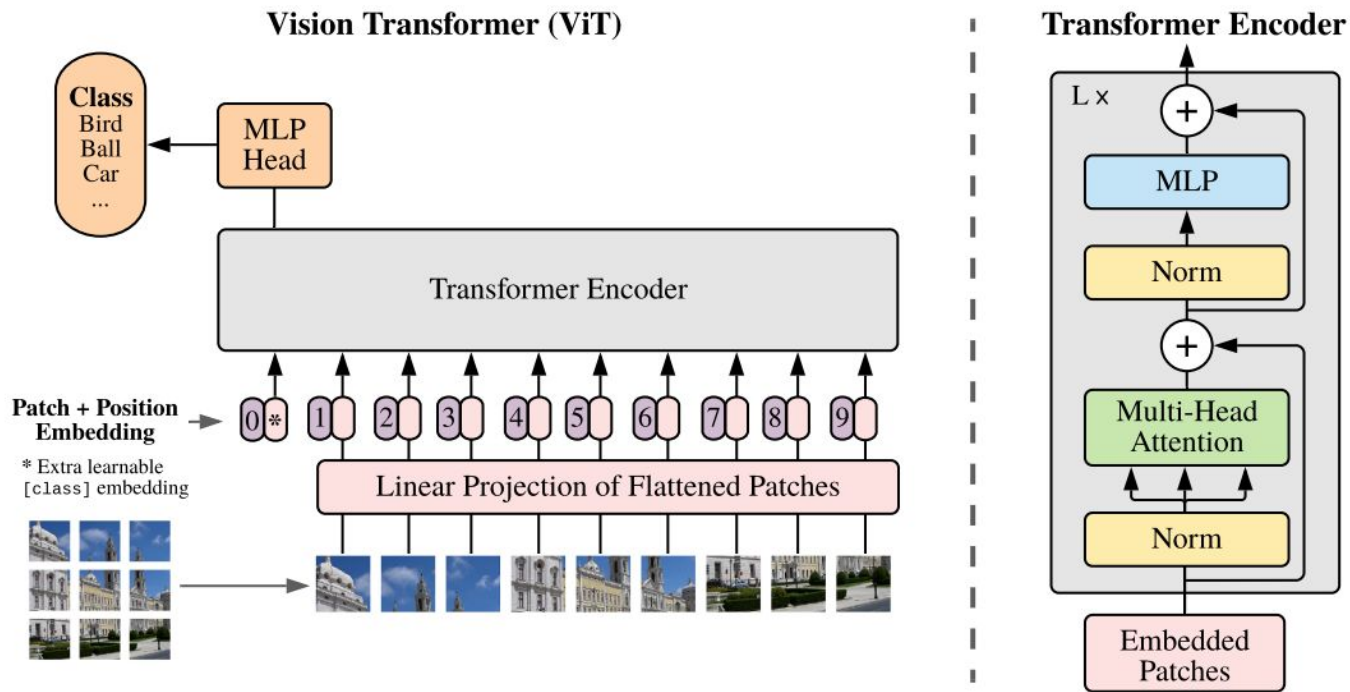
d
[?]

$d \rightarrow d+1$



$S_1: a b c$
 $S_2: b c a$

Vision Transformer





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